Finance and Risk Analytics Project Report

Data Science and Business Analytics - Part B

Great Learning

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1. Draw Stock Price Graph (Stock Price vs Time) for any 2 given stocks with inference & Dataset summary

Information about the dataset

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 314 entries, 0 to 313
Data columns (total 11 columns):
 #   Column               Non-Null Count  Dtype
---  ------               --------------  -----
 0   Date                 314 non-null    object
 1   Infosys              314 non-null    int64
 2   Indian Hotel         314 non-null    int64
 3   Mahindra & Mahindra  314 non-null    int64
 4   Axis Bank            314 non-null    int64
 5   SAIL                 314 non-null    int64
 6   Shree Cement         314 non-null    int64
 7   Sun Pharma           314 non-null    int64
 8   Jindal Steel         314 non-null    int64
 9   Idea Vodafone        314 non-null    int64
 10  Jet Airways          314 non-null    int64
dtypes: int64(10), object(1)
memory usage: 27.1+ KB
```

Shape of the data: 314 rows and 10 columns

Head of the dataset:

	Date	Infosys	Indian Hotel	Mahindra & Mahindra	Axis Bank	SAIL	Shree Cement	Sun Pharma	Jindal Steel	Idea Vodafone	Jet Airways
0	31-03-2014	264	69	455	263	68	5543	555	298	83	278
1	07-04-2014	257	68	458	276	70	5728	610	279	84	303
2	14-04-2014	254	68	454	270	68	5649	607	279	83	280
3	21-04-2014	253	68	488	283	68	5692	604	274	83	282
4	28-04-2014	256	65	482	282	63	5582	611	238	79	243

Summary of the dataset:

	Infosys	Indian Hotel	Mahindra & Mahindra	Axis Bank	SAIL	Shree Cement	Sun Pharma	Jindal Steel	Idea Vodafone	Jet Airways
count	314.000000	314.000000	314.000000	314.000000	314.000000	314.000000	314.000000	314.000000	314.000000	314.000000
mean	511.340764	114.560510	636.678344	540.742038	59.095541	14806.410828	633.468153	147.627389	53.713376	372.659236
std	135.952051	22.509732	102.879975	115.835569	15.810493	4288.275085	171.855893	65.879195	31.248985	202.262668
min	234.000000	64.000000	284.000000	263.000000	21.000000	5543.000000	338.000000	53.000000	3.000000	14.000000
25%	424.000000	96.000000	572.000000	470.500000	47.000000	10952.250000	478.500000	88.250000	25.250000	243.250000
50%	466.500000	115.000000	625.000000	528.000000	57.000000	16018.500000	614.000000	142.500000	53.000000	376.000000
75%	630.750000	134.000000	678.000000	605.250000	71.750000	17773.250000	785.000000	182.750000	82.000000	534.000000
max	810.000000	157.000000	956.000000	808.000000	104.000000	24806.000000	1089.000000	338.000000	117.000000	871.000000

1. **Scatter plot for Infosys and Jet Airways:**

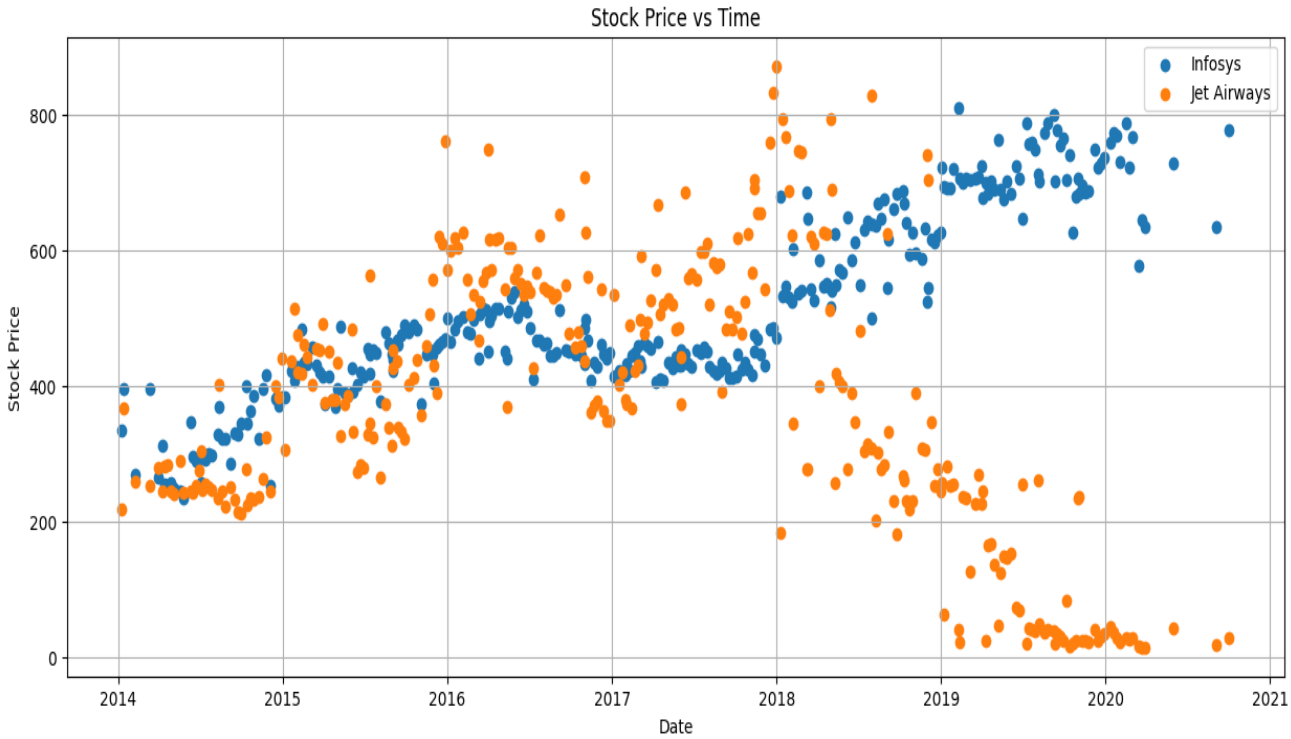

Figure 1: Scatter plot for Infosys and Jet Airways

As per the data its has been observed the there is a declination in the stocks for airlines company Jet airways due to many factors like covid and for IT based company grown during the covid.

Also, we will check the monthly yearly and quarterly for different companies

2. **Monthly stock price trends for Infosys and Indian Hotel**

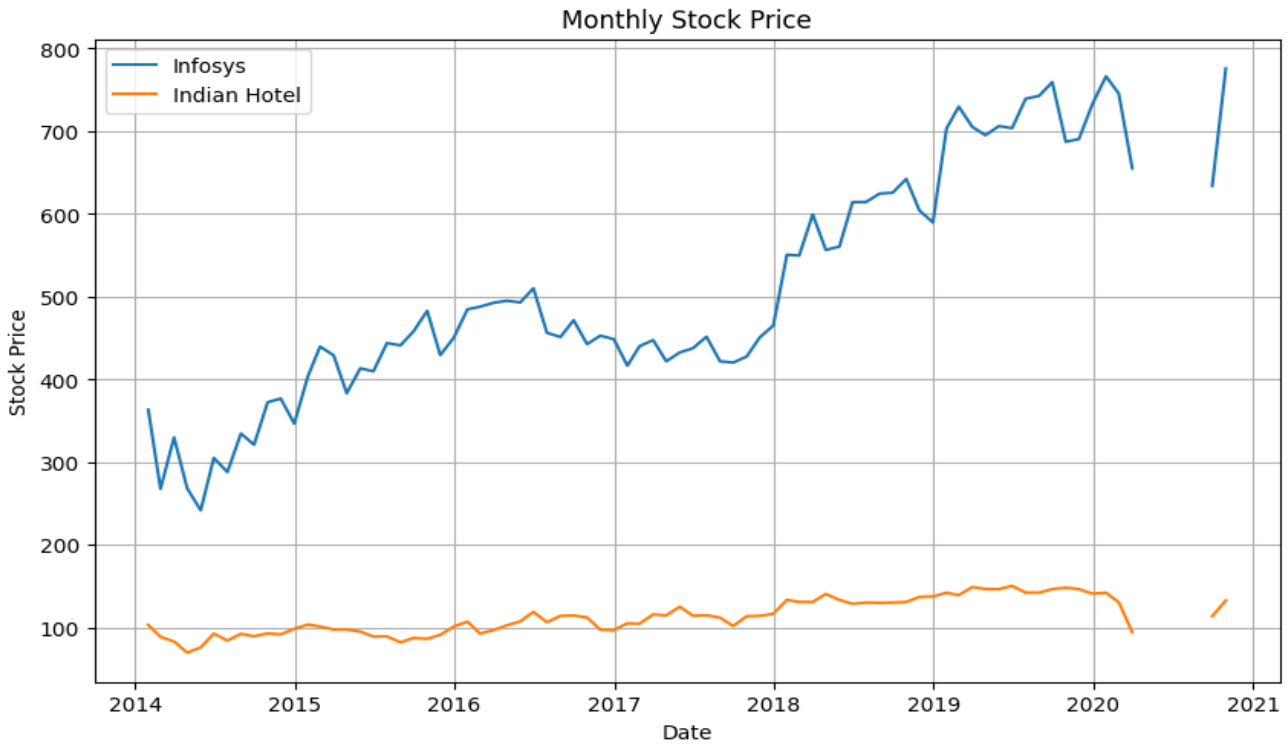

Figure 2: Monthly - Line plot for Infosys and Indian Hotel

3. Yearly stock price trends for Infosys and Indian Hotel

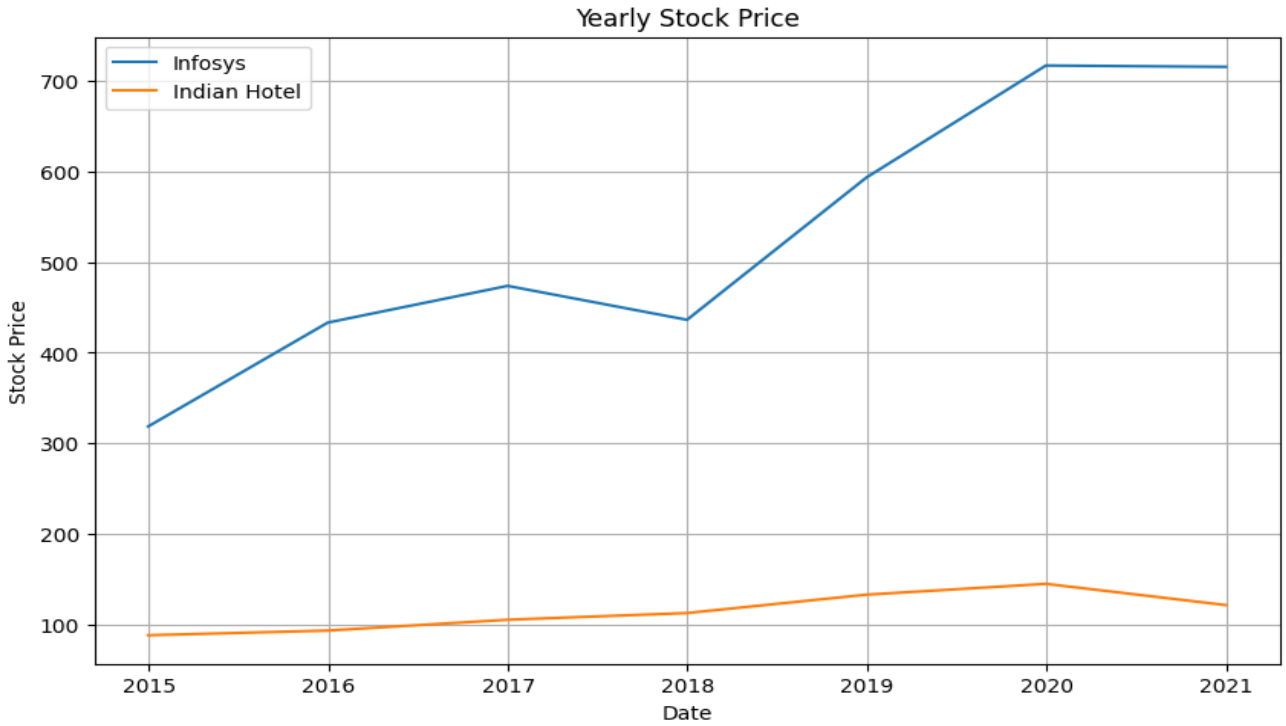

Figure 3: Yearly - Line plot for Infosys and Indian Hotel

4. Quarterly stock price trends for Infosys and Indian Hotel

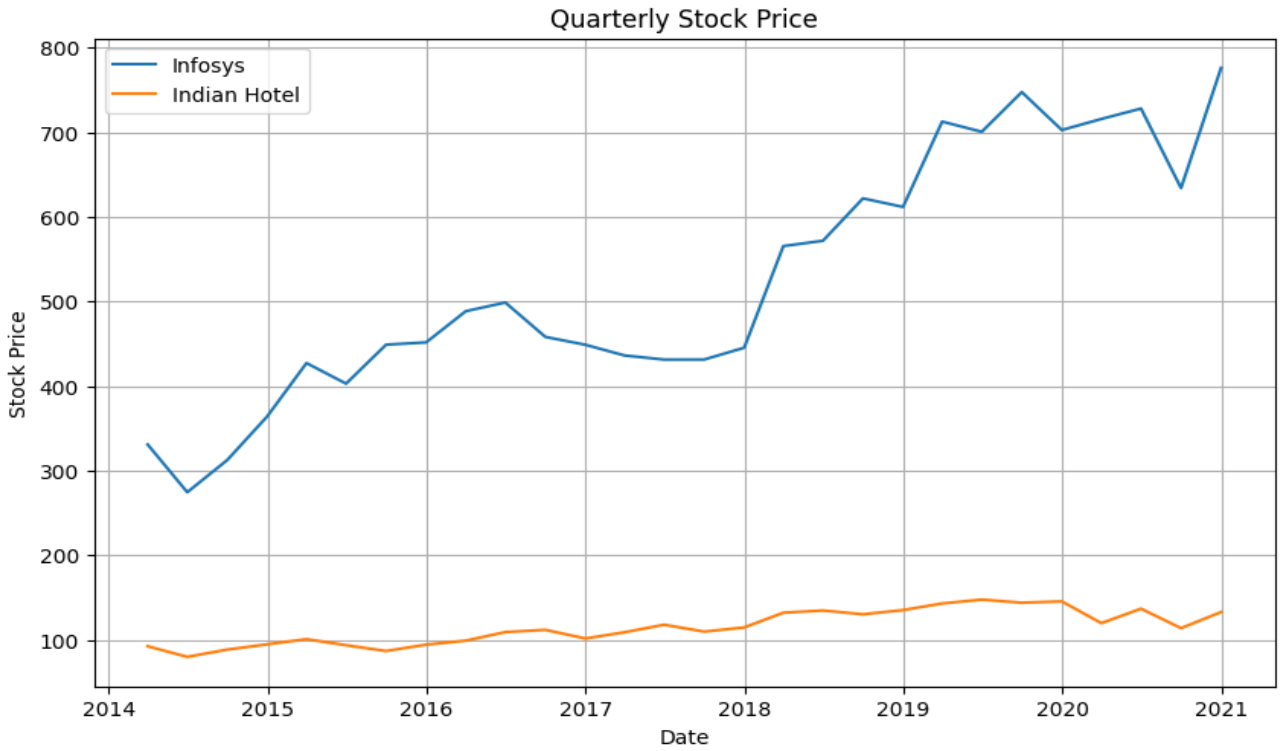

Figure 4: Yearly - Line plot for Infosys and Indian Hotel

2. Calculate Returns for all stocks with inference

Results:

```
              Infosys  Indian Hotel  Mahindra & Mahindra  Axis Bank      SAIL  \
Date
2014-03-31        NaN           NaN                  NaN        NaN       NaN
2014-07-04  -0.026873     -0.014599             0.006572   0.048247  0.028988
2014-04-14  -0.011742      0.000000            -0.008772  -0.021979 -0.028988
2014-04-21  -0.003945      0.000000             0.072218   0.047025  0.000000
2014-04-28   0.011788     -0.045120            -0.012371  -0.003540 -0.076373
...               ...           ...                  ...        ...       ...
2020-02-03   0.009649     -0.110348             0.030305  -0.057580 -0.087011
2020-09-03  -0.139625     -0.051293            -0.093819  -0.145324 -0.095310
2020-03-16  -0.094207     -0.236389            -0.285343  -0.284757 -0.105361
2020-03-23   0.109856     -0.182322            -0.091269  -0.173019 -0.251314
2020-03-30  -0.017228      0.000000            -0.031198   0.051432  0.090972

             Shree Cement  Sun Pharma  Jindal Steel  Idea Vodafone  Jet Airways
Date
2014-03-31           NaN         NaN           NaN            NaN          NaN
2014-07-04      0.032831    0.094491     -0.065882       0.011976     0.086112
2014-04-14     -0.013888   -0.004930      0.000000      -0.011976    -0.078943
2014-04-21      0.007583   -0.004955     -0.018084       0.000000     0.007117
2014-04-28     -0.019515    0.011523     -0.140857      -0.049393    -0.148846
...                  ...         ...           ...            ...          ...
2020-02-03      0.023688    0.072383     -0.053346      -0.287682    -0.127833
2020-09-03     -0.081183   -0.043319     -0.187816       0.693147    -0.200671
2020-03-16     -0.119709   -0.050745     -0.141830      -0.693147    -0.117783
2020-03-23     -0.067732   -0.076851     -0.165324       0.000000    -0.133531
2020-03-30     -0.006816    0.040585     -0.081917       0.000000     0.000000
```

Stock returns head:

Date	Infosys	Indian Hotel	Mahindra & Mahindra	Axis Bank	SAIL	Shree Cement	Sun Pharma	Jindal Steel	Idea Vodafone	Jet Airways
2014-03-31	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
2014-07-04	-0.026873	-0.014599	0.006572	0.048247	0.028988	0.032831	0.094491	-0.065882	0.011976	0.086112
2014-04-14	-0.011742	0.000000	-0.008772	-0.021979	-0.028988	-0.013888	-0.004930	0.000000	-0.011976	-0.078943
2014-04-21	-0.003945	0.000000	0.072218	0.047025	0.000000	0.007583	-0.004955	-0.018084	0.000000	0.007117
2014-04-28	0.011788	-0.045120	-0.012371	-0.003540	-0.076373	-0.019515	0.011523	-0.140857	-0.049393	-0.148846

Stock Returns Summary:

	Infosys	Indian Hotel	Mahindra & Mahindra	Axis Bank	SAIL	Shree Cement	Sun Pharma	Jindal Steel	Idea Vodafone	Jet Airways
count	313.000000	313.000000	313.000000	313.000000	313.000000	313.000000	313.000000	313.000000	313.000000	313.000000
mean	0.002794	0.000266	-0.001506	0.001167	-0.003463	0.003681	-0.001455	-0.004123	-0.010608	-0.009548
std	0.035070	0.047131	0.040169	0.045828	0.062188	0.039917	0.045033	0.075108	0.104315	0.097972
min	-0.167300	-0.236389	-0.285343	-0.284757	-0.251314	-0.129215	-0.179855	-0.283768	-0.693147	-0.458575
25%	-0.014514	-0.023530	-0.020884	-0.022473	-0.040822	-0.019546	-0.020699	-0.049700	-0.045120	-0.052644
50%	0.004376	0.000000	0.001526	0.001614	0.000000	0.003173	0.001530	0.000000	0.000000	-0.005780
75%	0.024553	0.027909	0.019894	0.028522	0.032790	0.029873	0.023257	0.037179	0.024391	0.036368
max	0.135666	0.199333	0.089407	0.127461	0.309005	0.152329	0.166604	0.243978	0.693147	0.300249

Overview: The provided data presents a snapshot of returns for various stocks over a specified period, offering insights into their performance and volatility.

- Minimum Returns: The minimum recorded returns across all stocks range from -69.31% to -45.86%, indicating the lowest performance observed during the analyzed timeframe.
- 25th Percentile (Q1): At the 25th percentile, returns range from -5.26% to -1.45%, signifying the bottom quartile of returns where 25% of the data falls, reflecting moderate to poor performance.
- Median Returns (50th Percentile): The median returns range from -0.58% to 0.44%, serving as the midpoint of the data, with half of the returns falling below this value, indicating the central tendency of the dataset.
- 75th Percentile (Q3): At the 75th percentile, returns range from 2.44% to 3.72%, representing the upper quartile where 75% of the data falls, highlighting relatively stronger performance compared to the lower quartile.
- Maximum Returns: The maximum recorded returns range from 12.75% to 69.31%, showcasing the highest performance observed during the analyzed period, potentially indicating significant growth or exceptional returns for certain stocks.

This summary provides a concise yet informative overview of the returns dataset, allowing stakeholders to understand the range and distribution of returns across different stocks.

3. Calculate Stock Means and Standard Deviation for all stocks with inference

```
Mean Returns:
Infosys                 0.002794
Indian Hotel            0.000266
Mahindra & Mahindra    -0.001506
Axis Bank               0.001167
SAIL                   -0.003463
Shree Cement            0.003681
Sun Pharma             -0.001455
Jindal Steel           -0.004123
Idea Vodafone          -0.010608
Jet Airways            -0.009548
dtype: float64
```

```
Standard Deviation of Returns:
Infosys                 0.035070
Indian Hotel            0.047131
Mahindra & Mahindra     0.040169
Axis Bank               0.045828
SAIL                    0.062188
Shree Cement            0.039917
Sun Pharma              0.045033
Jindal Steel            0.075108
Idea Vodafone           0.104315
Jet Airways             0.097972
dtype: float64
```

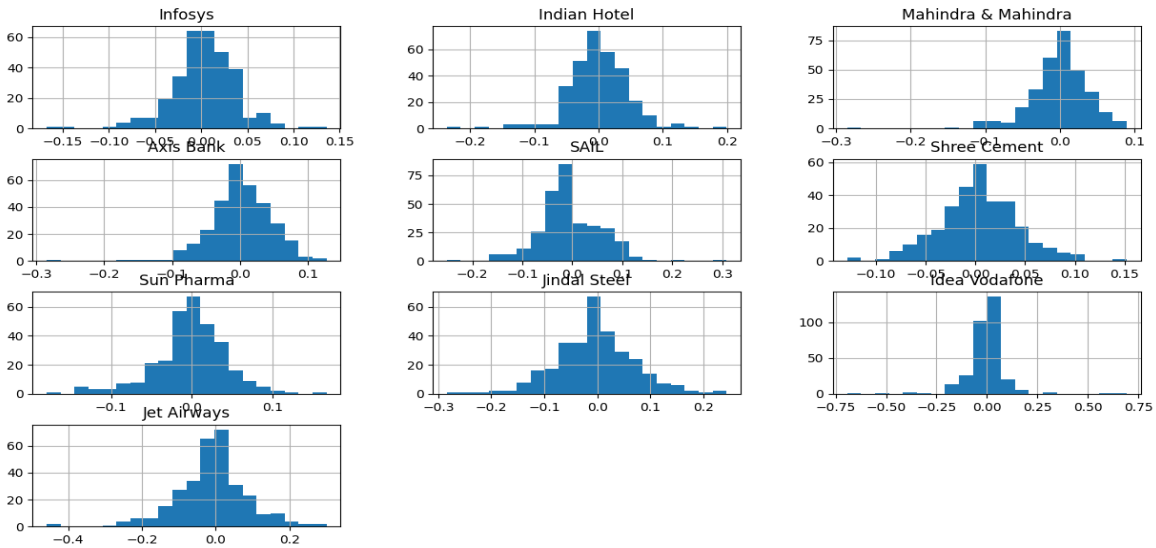

Figure 5: Histogram - Returns of each stock

Inferences: Mean returns

Infosys, Axis Bank, and Shree Cement exhibit positive mean returns, indicating overall positive performance.

Stocks like Indian Hotel, Mahindra & Mahindra, Sun Pharma, Jindal Steel, Idea Vodafone, and Jet Airways have relatively lower mean returns, suggesting weaker performance.

Standard Deviation of Returns:

Stocks with higher standard deviations like Jindal Steel, Idea Vodafone, and Jet Airways indicate greater volatility and risk.

Infosys, Indian Hotel, Mahindra & Mahindra, Axis Bank, and Shree Cement have relatively lower standard deviations, suggesting lower risk.

4. Draw a plot of Stock Means vs Standard Deviation and state your inference

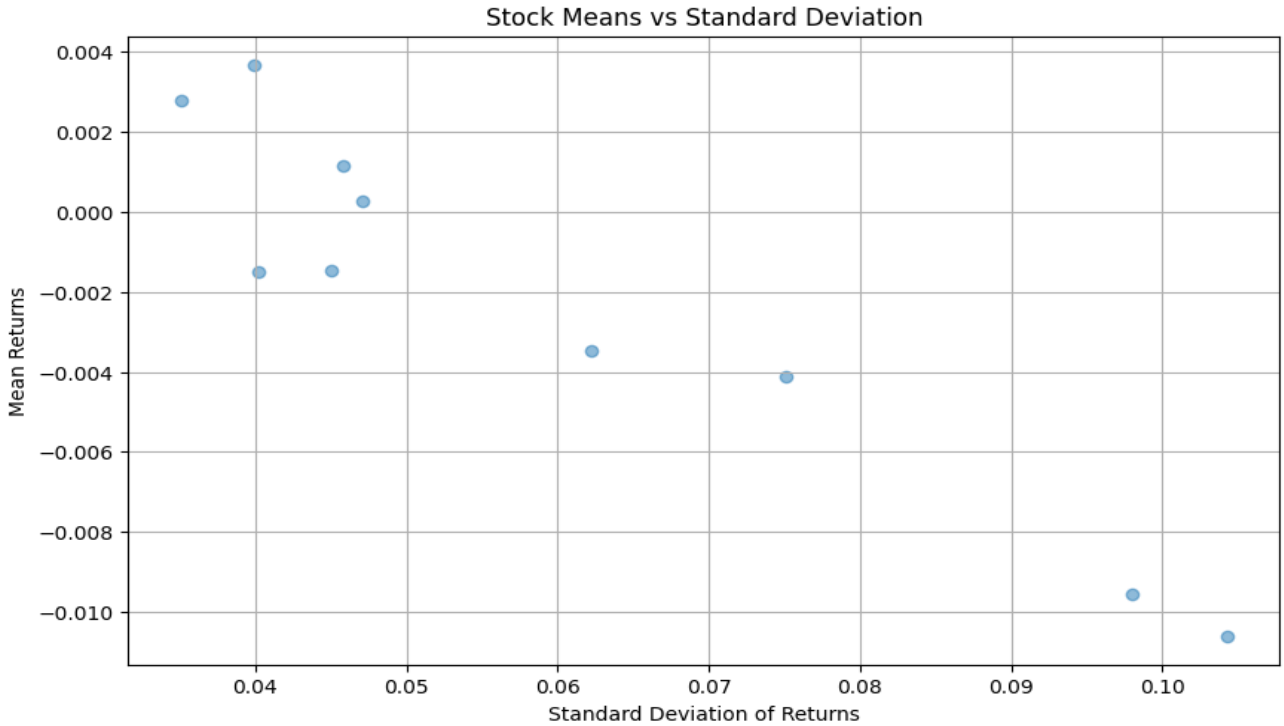

Figure 6: Scatter plot - Stock means vs Standard Deviation

The scatter plot illustrates the relationship between mean returns and standard deviation of returns for various stocks, showcasing the trade-off between risk and potential reward in the stock market.

Stocks with lower mean returns and higher standard deviation may not be favorable choices for inclusion in a portfolio, especially when there are other stocks available with higher returns and lower risk profiles.

5. Conclusions and Recommendations

	Average	Volatility
Infosys	0.002794	0.035070
Indian Hotel	0.000266	0.047131
Mahindra & Mahindra	-0.001506	0.040169
Axis Bank	0.001167	0.045828
SAIL	-0.003463	0.062188
Shree Cement	0.003681	0.039917
Sun Pharma	-0.001455	0.045033
Jindal Steel	-0.004123	0.075108
Idea Vodafone	-0.010608	0.104315
Jet Airways	-0.009548	0.097972

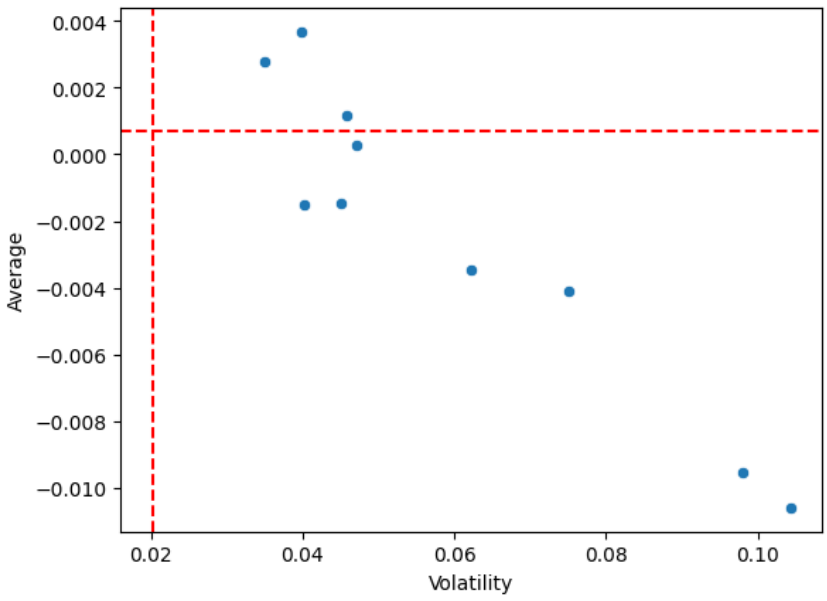

Figure 7: Scatter plot – Volatility vs Average

Diversity of Returns: The average returns across the stocks vary significantly, ranging from positive values (e.g., Infosys, Axis Bank, Shree Cement) to negative values (e.g., SAIL, Jindal Steel, Idea Vodafone, Jet Airways). This indicates a diverse range of performance among the stocks.

Risk Levels: Stocks with higher volatility tend to have higher risk levels. Stocks like Idea Vodafone and Jet Airways exhibit notably higher volatility compared to others, suggesting higher risk associated with investing in these stocks.

Stability: Some stocks, such as Shree Cement and Infosys, have relatively lower volatility coupled with positive average returns. These stocks may be perceived as relatively stable investments compared to others in the dataset.

Potential Opportunities: Stocks with lower average returns but lower volatility, such as Indian Hotel and Mahindra & Mahindra, may present opportunities for investors seeking a balance between returns and risk.

Risk Management: Understanding the relationship between average returns and volatility is crucial for effective risk management. Investors may need to diversify their portfolios to include a mix of stocks with varying levels of volatility to mitigate overall portfolio risk.

Overall, analyzing both average returns and volatility provides valuable insights for investors to make informed decisions regarding portfolio composition, risk tolerance, and investment strategies.

Performance Ranking:

Among the analyzed stocks, Infosys, Axis Bank, and Shree Cement demonstrate comparatively better performance with positive mean returns.

Stocks with lower mean returns, such as Indian Hotel, Mahindra & Mahindra, Sun Pharma, Jindal Steel, Idea Vodafone, and Jet Airways, may require closer scrutiny for potential improvement strategies.

Risk Assessment:

Investors seeking lower risk may consider allocating funds to stocks with lower standard deviations like Infosys, Indian Hotel, Mahindra & Mahindra, Axis Bank, and Shree Cement.

Stocks with higher standard deviations such as Jindal Steel, Idea Vodafone, and Jet Airways should be approached with caution due to their higher volatility.

Portfolio Diversification:

Diversification across stocks with varying risk levels can help mitigate overall portfolio risk while potentially capitalizing on diverse return opportunities.

Balancing high-risk, high-return stocks with lower-risk, stable performers can optimize portfolio performance while managing risk exposure.

This summary and conclusion provide insights into the performance and risk characteristics of the analyzed stocks, aiding investors in making informed decisions regarding portfolio composition and risk management strategies.